Regression for Regulatory Genomics

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Learning objectives

By the end of this lecture, you will be able to:

1. Apply regression methods to genomic data integration problems
2. Understand fine-mapping and variable selection in regulatory genomics
3. Integrate GWAS and regulatory data using TWAS and colocalization

Roadmap: From biology to methodology

Lecture 12 (Biology)

- Regulatory genomics data types
- ChIP-seq, ATAC-seq, DNA methylation
- Biological interpretation

Today (Methodology)

- Polygenic score model
- eQTLs and chromatin accessibility
- Regression for genomic integration
- Integrative analysis methods

We will discuss multivariate linear regression

Genetic data as an anchor variable

Genetic variants $\rightarrow$ Molecular Phenotype $\rightarrow$ Disease Phenotype

Using 1000 Genomes data via bigsnpr

```r
library(bigsnpr)
download_1000G("../data/genotype/")
```

```r
data <- snp_attach(.bk.file)
str(data, max.level=1, strict.width = "cut")
```

```
## List of 3
##  $ genotypes:Reference class 'FBM.code256' [package "bigstatsr"] with 16 fields
##   ..and 26 methods, of which 12 are  possibly relevant
##  $ fam      :'data.frame':   2490 obs. of  6 variables:
##  $ map      :'data.frame':   1664852 obs. of  6 variables:
##  - attr(*, "class")= chr "bigSNP"
```

Today's lecture

1. Polygenic risk prediction (G $\rightarrow$ phenotype)
2. Fundamental challenges in PGS (regression)
3. Expression Quantitative Trait Loci (G $\rightarrow$ Expr.)
4. Co-localization in genomics

Recap: Notations

- X: $n \times p$ genotype matrix of n samples/individuals and p variants
- Y: $n \times 1$ phenotype vector of n samples/individuals

Let's do GWAS to find the disease-associated variants

A variant-by-variant association t-test for a variant j:

$$
\begin{aligned}
H_0 : &\quad \mathbb{E}[X_{ij}|Y_i = 1] = \mathbb{E}[X_{ij}|Y_i = 0] \\
&\quad \text{vs.} \\
H_1 : &\quad \mathbb{E}[X_{ij}|Y_i = 1] \neq \mathbb{E}[X_{ij}|Y_i = 0]
\end{aligned}
$$

- The average genotype is the same between the case and control under the null.

GWAS statistics over all the variants

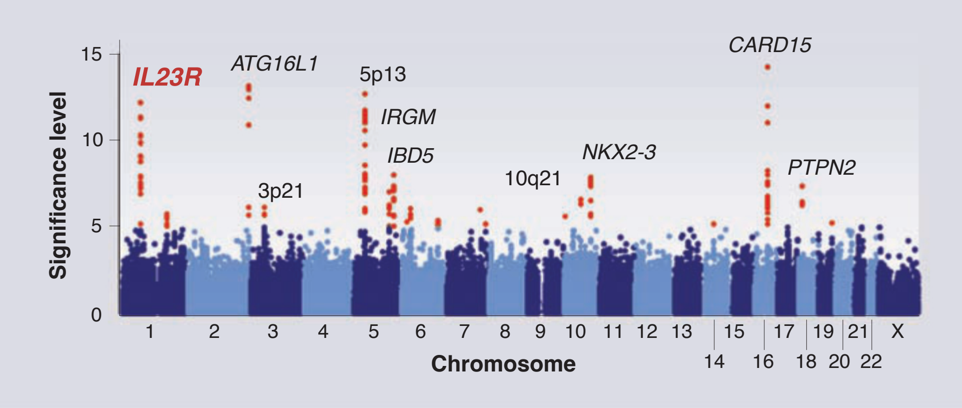
```r
## library(matrixTests)
.gwas <- col_t_welch(X[Y == 0, ], X[Y == 1, ])
```

- For each pair of columns in the two matrices, the function performs a t-test and summarizes all the results into a list of vectors.

```r
names(.gwas)
```

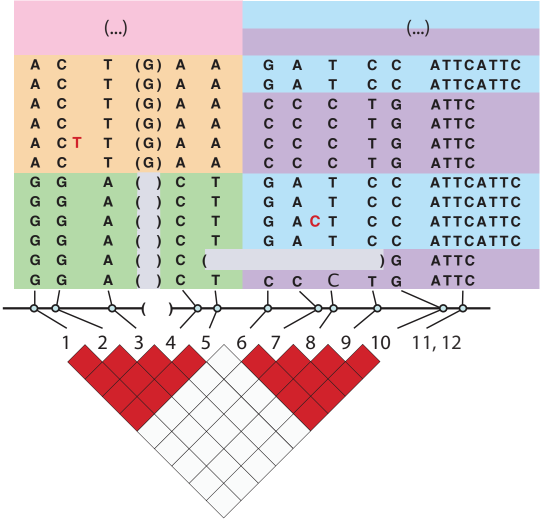
```
##  [1] "obs.x"       "obs.y"       "obs.tot"     "mean.x"      "mean.y"
##  [6] "mean.diff"   "var.x"       "var.y"       "stderr"      "df"
## [11] "statistic"   "pvalue"      "conf.low"    "conf.high"   "mean.null"
## [16] "alternative" "conf.level"
```

Human genetic variation recap



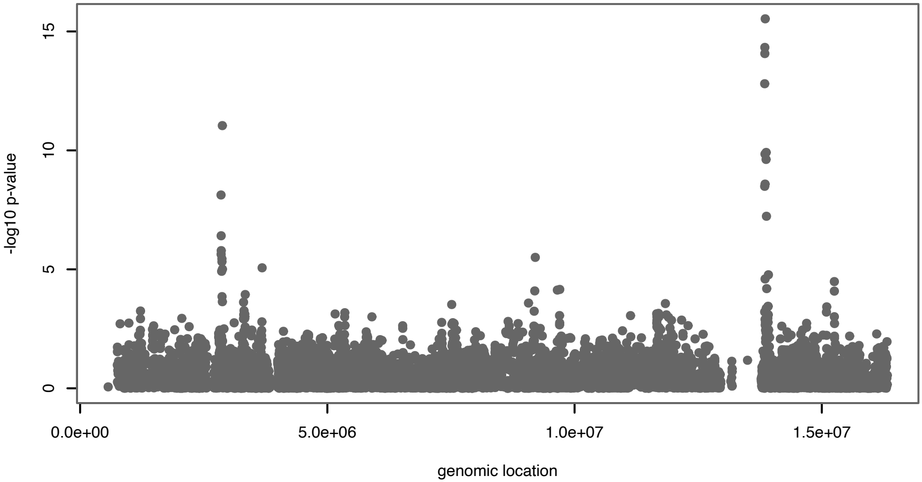
(Altshuler et al. 2008)

Human genetic variation recap

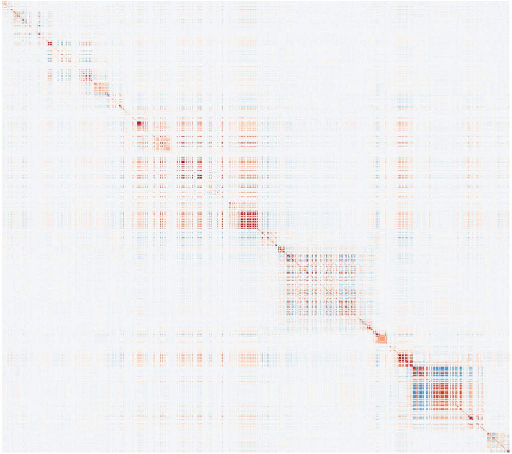


(Altshuler et al. 2008)

Manhattan plot: a summary of all the GWAS p-values



The covariance of X between the variants (columns)



For a genotype matrix X $(n \times p)$,

Linkage disequilibrium matrix

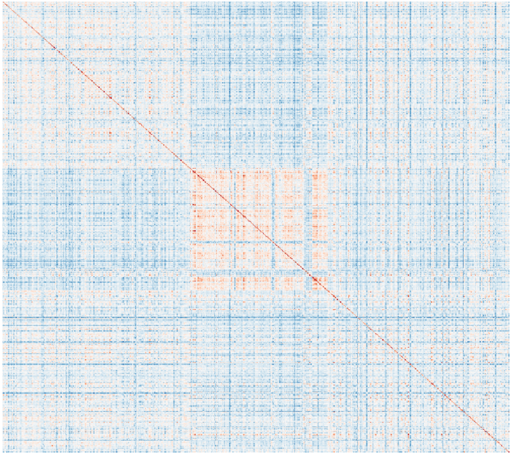
A $p \times p$ matrix:

$$\hat{R} = \frac{1}{n} X^{\top} X$$

column $\mathbf{x}_j$ standardized (mean 0, SD 1).

$$R_{ij} = \frac{1}{n} \sum_{r=1}^{n} X_{ri} X_{rj}$$

The covariance of X between the individuals (rows)



For a genotype matrix X $(n \times p)$

Genetic relatedness matrix

An $n \times n$ matrix:

$$\hat{K} = \frac{1}{p} X X^{\top}$$

each row $\mathbf{x}_i$ standardized (mean 0, SD 1).

$$K_{ij} = \frac{1}{p} \sum_{g=1}^{p} X_{ig} X_{jg}$$

What's next after GWAS?

- GWAS until 2010s: heavy focuses on **mapping**
  - GWAS map: genetic variants $\rightarrow$ a phenotype
  - Stringent genome-wide p-value cutoff
  - Study design, meta analysis
- NHGRI-EBI GWAS Catalog: `https://www.ebi.ac.uk/gwas/`

Let's take a look at the GWAS Catalog: `https://www.ebi.ac.uk/gwas/`

What's next after GWAS?

- GWAS since 2010s: more emphasis on **prediction**
  - Can we turn GWAS results to a prediction model?
  - Can we understand the mechanisms?
  - Machine learning, data integration, causal inference

Polygenic score models predict disease prevalence based on genome

- Poly + genic = many genes (units of hereditary components)

A (linear) polygenic score (PGS)

$$Y_i = \sum_j X_{ij} \beta_j$$

for an individual i.

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A (linear) polygenic score (PGS)

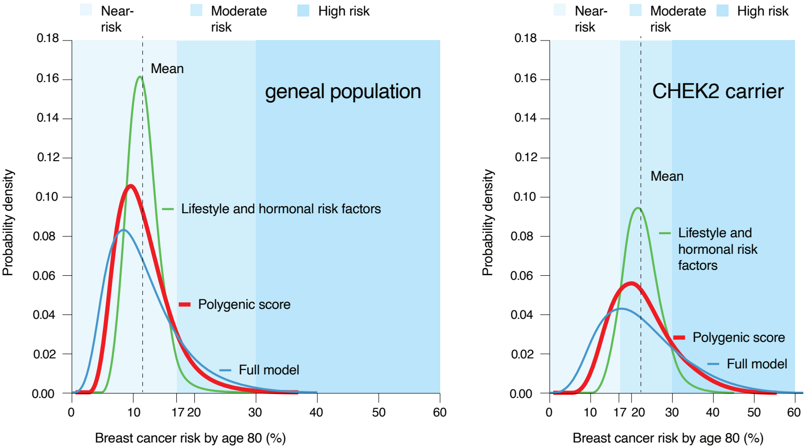
$$Y_i = \sum_j X_{ij}\beta_j$$

for an individual i.

Knowing β's, we can predict:

$$Y^{\star}{}_i \leftarrow \sum_j X^{\star}_{ij}\hat{\beta}_j$$

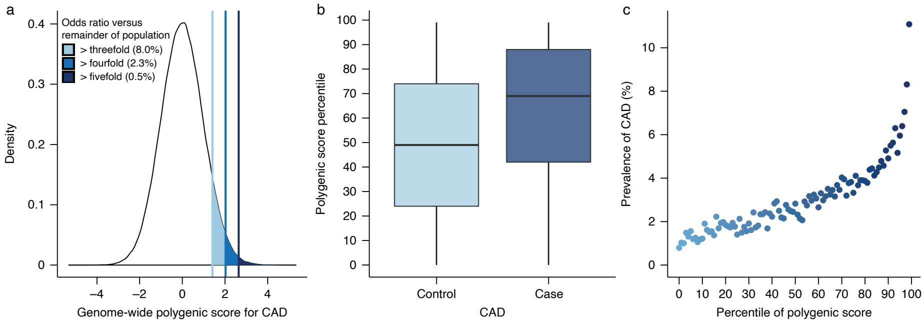
Example: PGS for breast cancer occurrence



- Strong heritability (proportion of disease risk variation explained by genetics): 35% - 80%
- Prediction is more accurate than lifestyle and other environmental factors
- More powerful if combined with rare risk factors

(Yang et al. 2023)

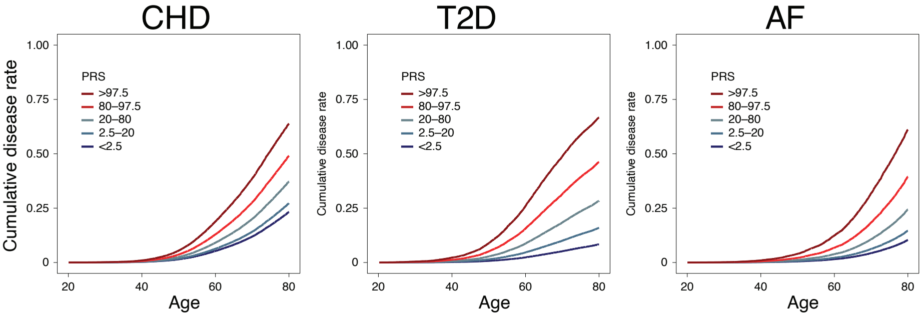
Example: Coronary artery disorder → prevention



- Top .5% of PGS values → five fold increase of CAD

(Khera et al. 2018)

Example: PGS stratifies individuals' disease onset

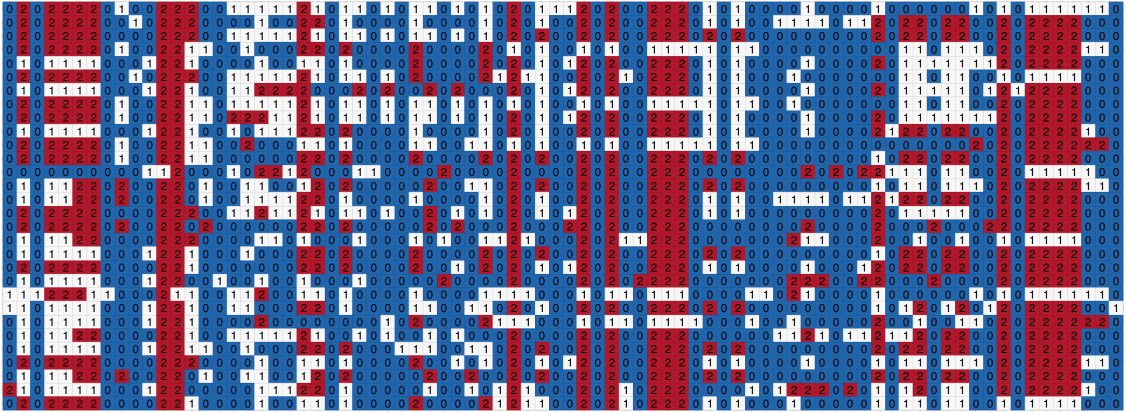


- PRS ≈ PGS.
- We can partition cohorts based on PGS profiles.

(Mars et al. 2020)

GWAS for “Obsessive `ggplot` Disorder”

A genotype matrix X ($X_{ij} \in \{0, 1, 2\}$):



[1] just for illustration purposes

In GWAS, we can have case vs. control phenotype

$Y_i = 1$ if case vs. $Y_i = 0$ if control

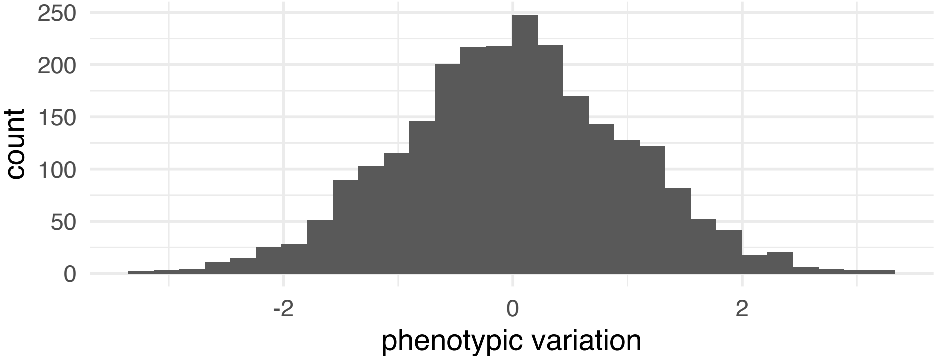
For our *OGD* GWAS:

```r
table(y)
```

```
## y
##    0   1
## 731 769
```

- E.g., Cancer vs. non-cancer, schizophrenia vs. no mental disorder
- The fundamental question is, "Which individuals/subjects can be truly labelled control/wild type?"

Under the hood, there are quantitative “risk” scores



- E.g., height, BMI, parents’ age of death, how many packs of cigarettes, etc.

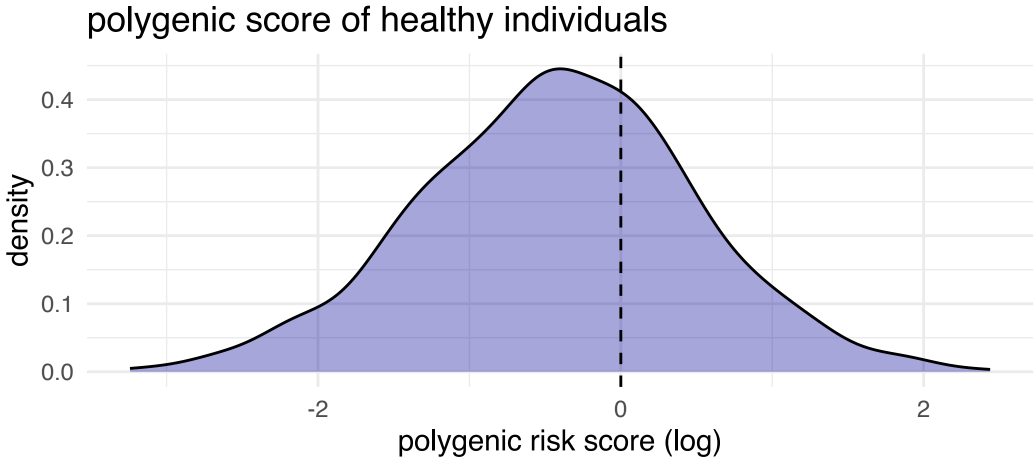
Side note: Almost all complex traits are polygenic

Polygenic effects of common SNPs $\rightarrow$ polymorphism in phenotypes

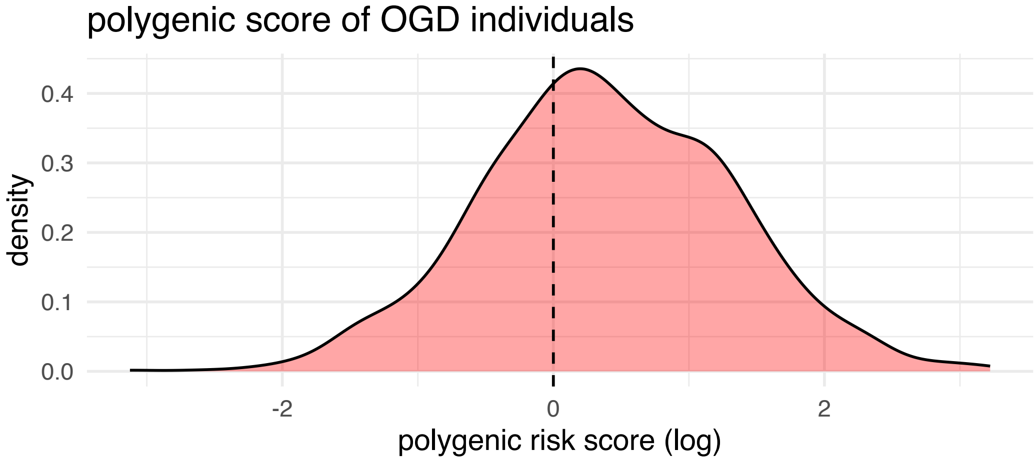
> *A polygenic trait is a characteristic, such as height or skin color, that is influenced by two or more genes.* ***Because multiple genes are involved, polygenic traits do not follow the patterns of Mendelian inheritance.*** *Many polygenic traits are also influenced by the environment and are called multifactorial.*

https://www.genome.gov/genetics-glossary/Polygenic-Trait

A polygenic score to predict disease prevalence



A polygenic score to predict disease prevalence



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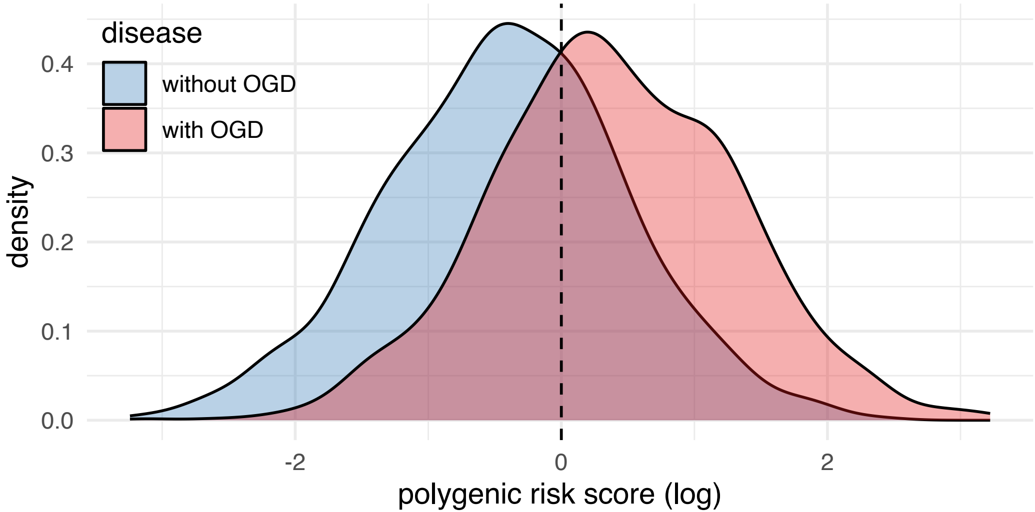

Polygenic score $\propto$ the odds of the case vs. control

Log-odds ratio:

$$g(y) = \log \frac{p(\text{disease}|Y > y)}{p(\text{no disease}|Y > y)}$$

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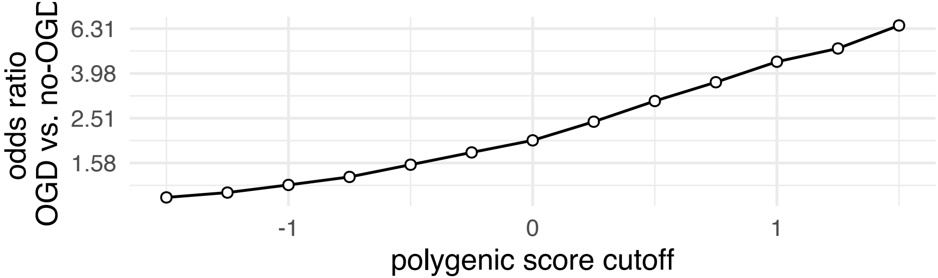

A PGS model to explain disease prevalence

Log-odds ratio:

$$g(y) = \log \frac{p(\text{disease}|Y > y)}{p(\text{no disease}|Y > y)}$$

Goal: Estimate a function to predict this log-odds ratio.

$$g(y_i) \sim \beta_1 X_{i1} + \beta_2 X_{i2} \cdots + \epsilon$$

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$$g(y_i) \sim \underbrace{\beta_1 X_{i1} + \beta_2 X_{i2} \cdots}_{\textbf{genetic effects}} + \epsilon$$

PGS estimation is a linear modelling with $p \gg n$

$$Y_i \sim X_{i1}\beta_1 + X_{i2}\beta_2 + \cdots$$

PGS estimation is a linear modelling with $p \gg n$

$$\begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix} \sim \begin{pmatrix} X_{11} \\ X_{21} \\ \vdots \\ X_{n1} \end{pmatrix} \beta_1 + \begin{pmatrix} X_{12} \\ X_{22} \\ \vdots \\ X_{n2} \end{pmatrix} \beta_2 + \cdots$$

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$$\mathbf{y} \sim \mathbf{x}_1\beta_1 + \mathbf{x}_2\beta_2 + \cdots + \mathbf{x}_p\beta_p$$

$$p \gg n$$

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$$\mathbf{y} \sim \mathbf{x}_1\beta_1 + \mathbf{x}_2\beta_2 + \cdots + \mathbf{x}_p\beta_p$$

$$p \gg n$$

$n \approx 10^4$ but $p \approx 10^6$ for many large-scale GWAS

$p \gg n$: Why can't we just fit a model?

```
## lm.out <- lm(Y ~ X)
```

- $p \gg n$: need to estimate $\beta_1, \ldots, \beta_p$

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```
## lm.out <- lm(Y ~ X)
```

- $p \gg n$: need to estimate $\beta_1, \ldots, \beta_p$
- How many samples? How many unknowns?
- Is it computationally feasible?

$p \gg n$: Variant-by-variant GWAS is not a bad idea

```
.gwas <- col_t_welch(X[Y == 0, ], X[Y == 1, ]) # univar.
```

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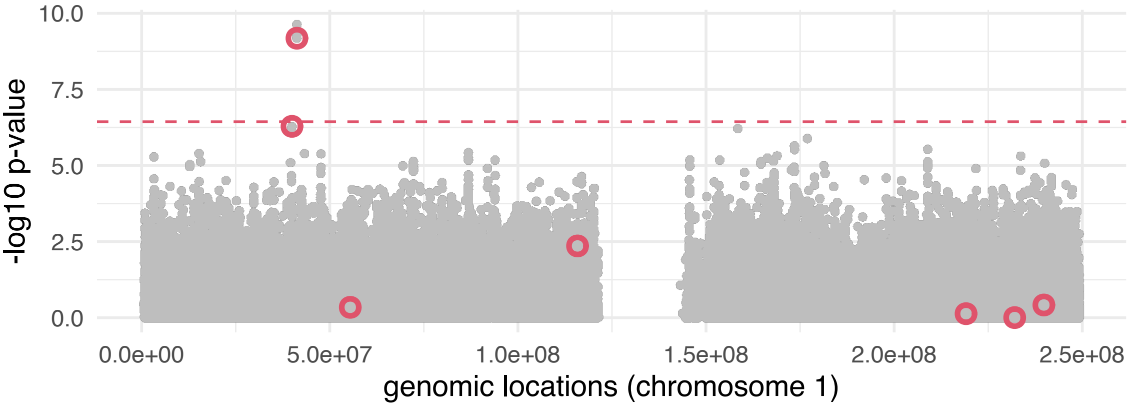


A linear model including only these “GWAS” variants

```r
X.gwas <- X[, p.adjust(.gwas$pvalue) < .05]
head(X.gwas, 3)
```

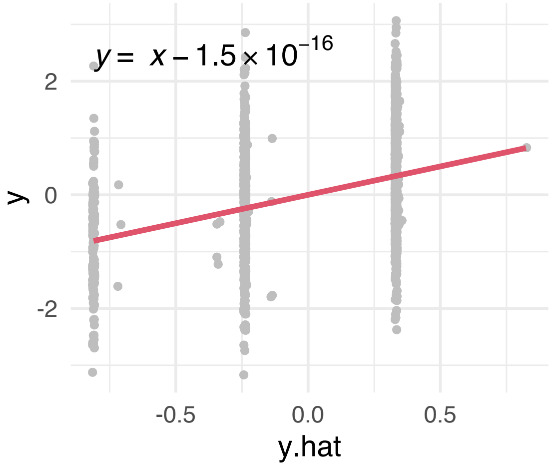
```
##          41270163 41270937 41272533 41274422
## HG00159        0        0        0        0
## HG01495        1        1        1        1
## HG01519        2        2        2        2
```

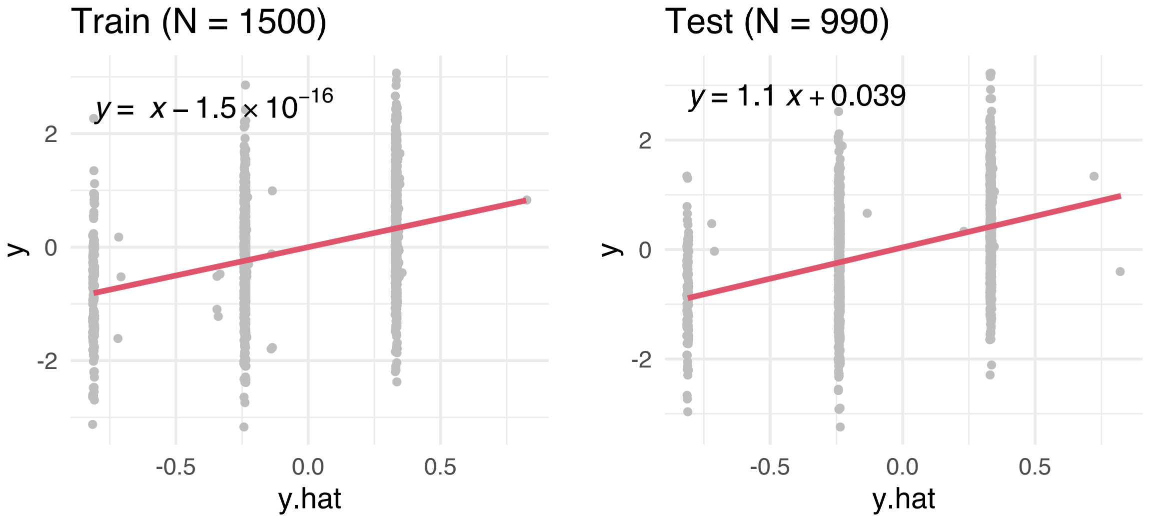
```r
lm.out <- lm(Y ~ X.gwas)
```

```r
anova(lm.out)
```

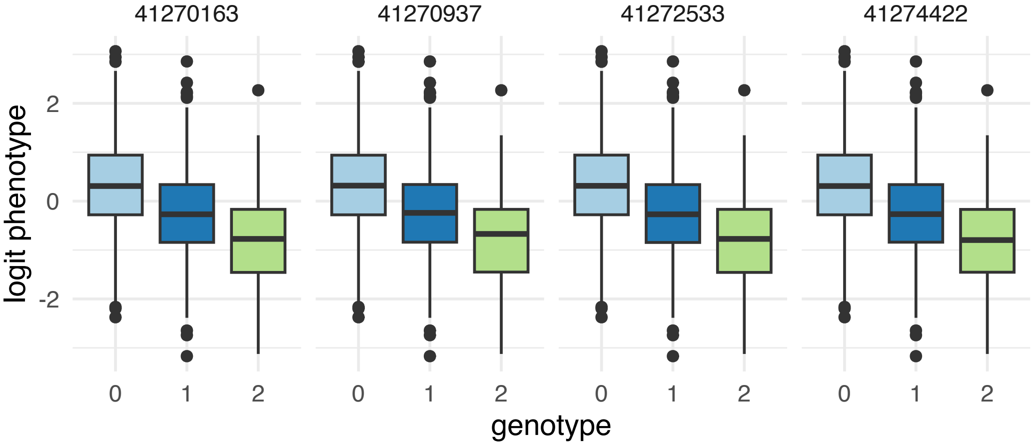
```
## Analysis of Variance Table
##
## Response: Y
##             Df  Sum Sq Mean Sq F value    Pr(>F)
## X.gwas       4  212.77  53.194  63.965 < 2.2e-16 ***
## Residuals 1495 1243.25   0.832
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Train (N = 1500)

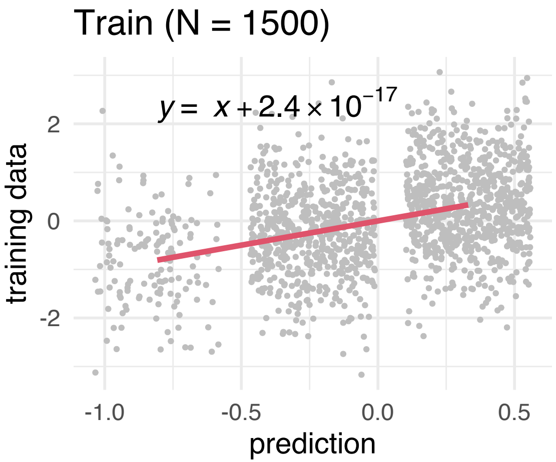


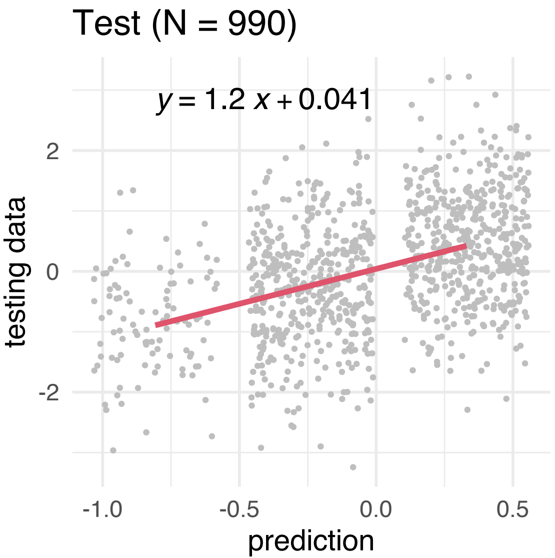

Train (N = 1500)
y = x − 1.5 × 10^−16
2
0
-2
y
-0.5
0.0
0.5
y.hat
Test (N = 990)
y = 1.1 x + 0.039
2
0
-2
y
-0.5
0.0
0.5
y.hat

Do we need all of these GWAS variants?



Just one of the GWAS variants works fine





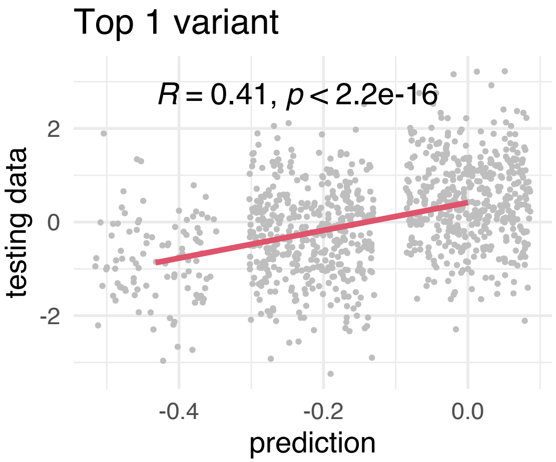
Simply add more variables?

Let's do some experiments:

1. Pick top K variants (ordering by GWAS, $p_{(1)} < p_{(2)} < \ldots$)
2. Take GWAS effect sizes β (the mean difference between case vs. control)
3. Predict by a linear combination of these effects:

$$\sum_{j=1}^{K} X_{i(j)} \beta_{(j)}$$

PGS is a variable selection problem



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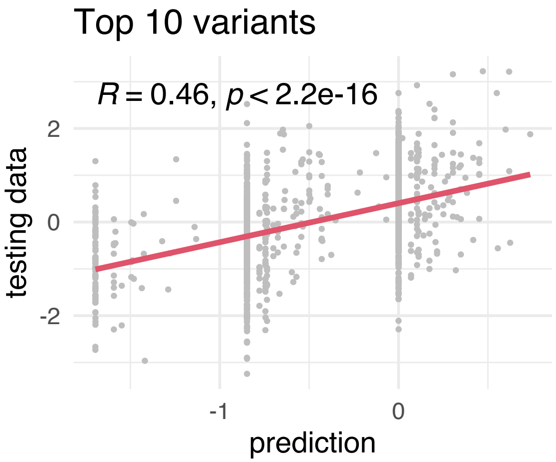

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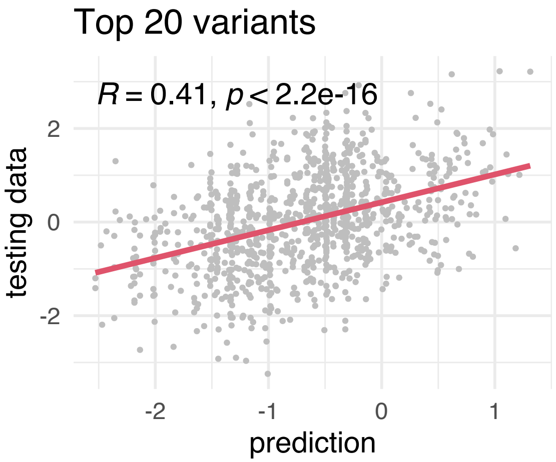

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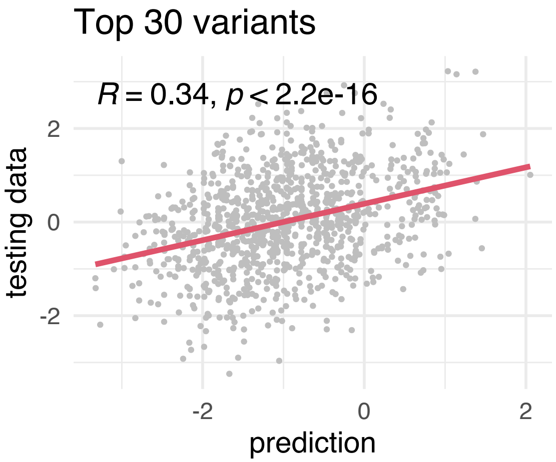

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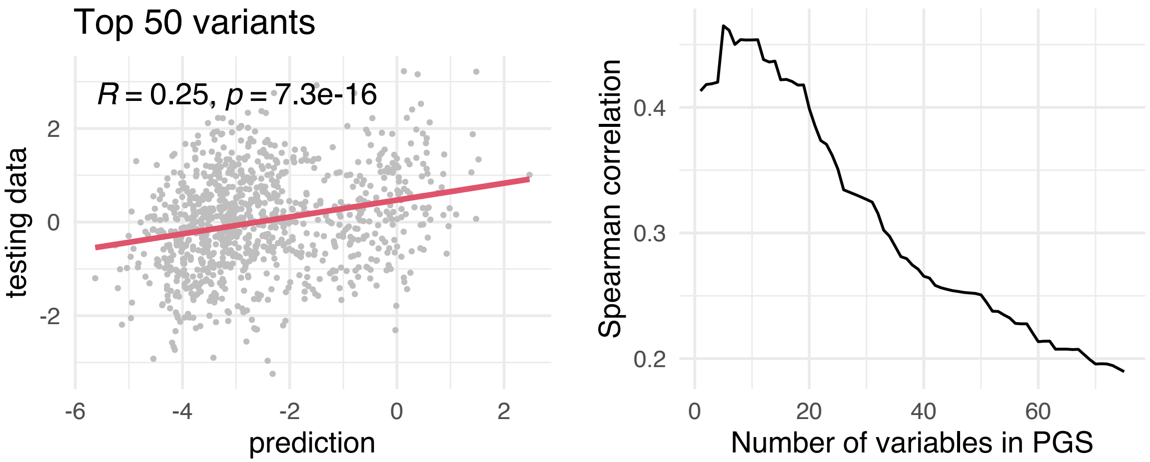

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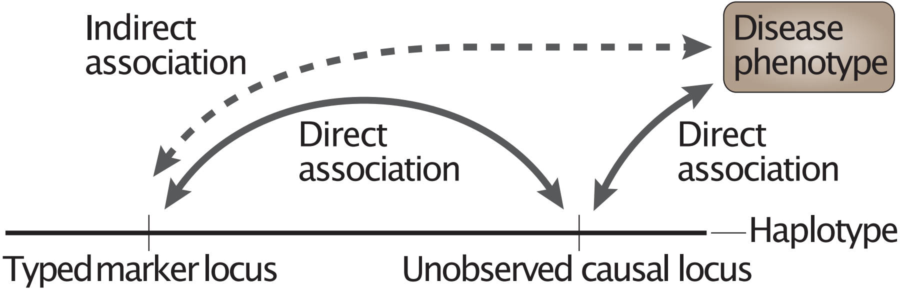
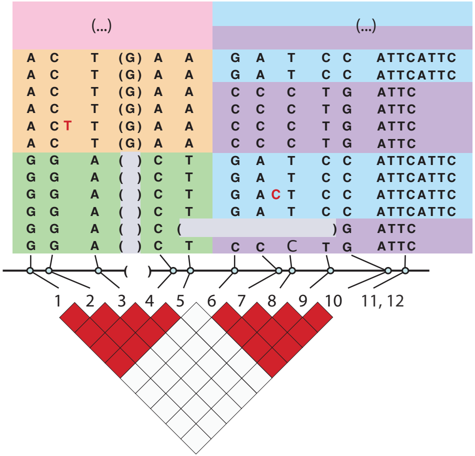
GWAS statistics only roughly tag causal variants



(Balding 2006)

Covariance between variants obfuscates GWAS interpretation

- Linkage disequilibrium (LD) block
- The result of recombination events throughout generations



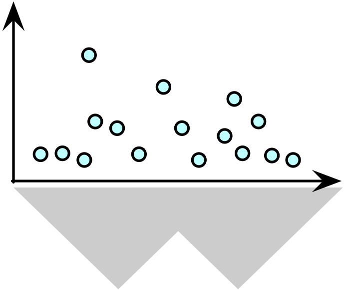
A typical setting of PGS training

PGS training

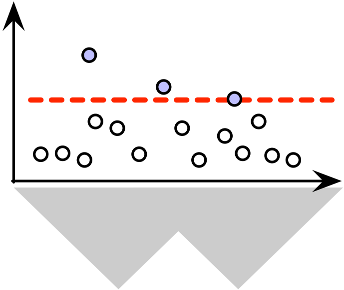
- Input:
  1. GWAS summary statistics (p-values, univariate effect sizes)
  2. $n \times p$ genotype matrix X, where $X_{ij} \in \{0, 1, 2\}$
  3. $n \times 1$ phenotype vector $\mathbf{y}$
- Output:
  - *Polygenic* effect sizes $\hat{\beta}_1, \ldots, \hat{\beta}_p$
- Objective:
  - We want to predict unseen $Y_i^\star$ by $f(X^\star; \hat{\beta})$ as accurately as possible
  - We want $\beta_j = 0$ as many as possible; only some $\beta_j \neq 0$.

Common strategies to deal with LD structures

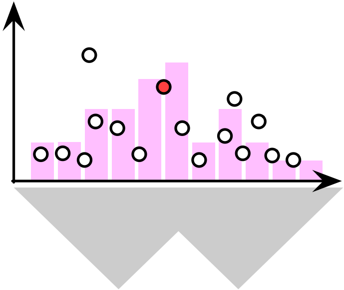
Strategy 1. Just use them all



Strategy 2. Pruning/clumping

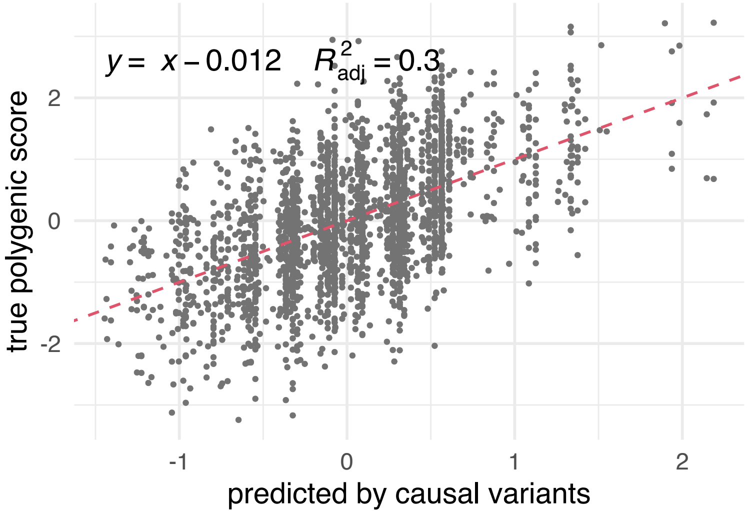


Strategy 3. **Fine-mapping**

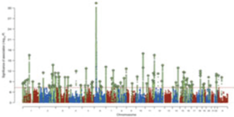


Causal variables explain a large fraction of var.

```r
.lm <- lm(.sim$y.q ~ .sim$x[, .sim$causal, drop = FALSE] - 1)
```



A general workflow of PGS estimation



Base data

↔ Independent samples

Target data

Data

- Summary statistics
- Betas/ORs weights in PRS calculation

- Individual-level genotype and phenotype data
- Often small sample size

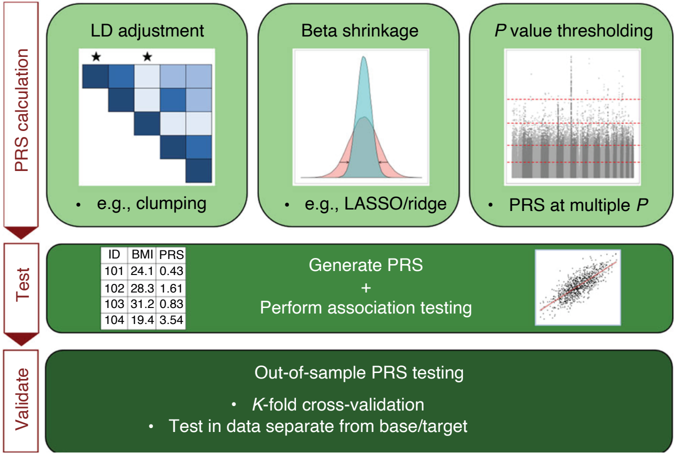
Processing

QC

- Both data sets QCed as standard in GWAS
- Some QC requires special care in PRS (e.g., sample overlap, relatedness and population structure)
- Retain set of SNPs that overlap between base and target data

(Choi et al. 2020)

A general workflow of PGS estimation



(Choi et al. 2020)

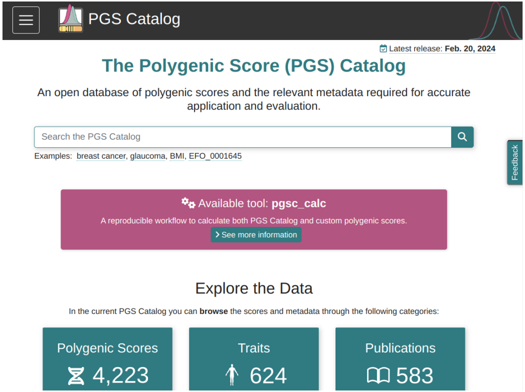
A proposed workflow of PGS estimation

1. Take GWAS summary statistics and target population genotype X
2. Run SuSiE[2] chromosome by chromosome to select variables
3. Predict PGS per chromosome and aggregate them

[2] SuSiE can run with summary statistics only

PGS Catalog: No need to train from scratch

https://www.pgscatalog.org/

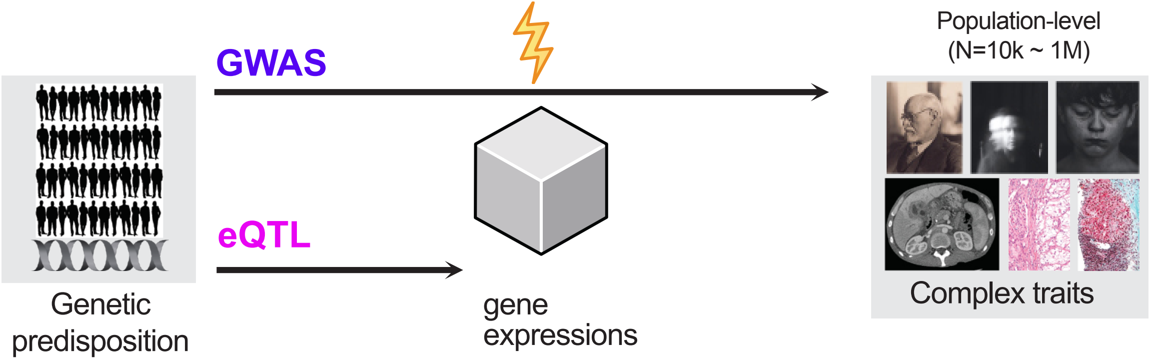


Today's lecture

1. Polygenic risk prediction (G $\rightarrow$ phenotype)
2. Fundamental challenges in PGS (regression)
3. Expression Quantitative Trait Loci (G $\rightarrow$ Expr.)
4. Co-localization in genomics

Transcriptome-wide association study (TWAS)

Using eQTL, we can understand GWAS mechanisms… (Gamazon et al. 2015)



$$\mathbf{m}_g \sim X\alpha_g + \epsilon_g \quad \Longrightarrow \quad \mathbf{y}_{\text{GWAS}} = \sum_{g \in \text{causal genes}} \mathbf{m}_g \beta_g + \epsilon_y$$

eQTL mapping as a linear regression problem

A variant-by-variant association test for a variant j:

$$
\begin{aligned}
H_0 : &\quad \mathbb{E}[Y_i | X_{ij}] = \mu \\
&\quad \text{vs.} \\
H_1 : &\quad \mathbb{E}[Y_i | X_{ij}] = \mu + X_{ij}\beta_j
\end{aligned}
$$

- For gene expression Y, test association with each variant X_j
- Model: $Y_i = X_{ij}\beta_j + \epsilon$
- Repeat for all gene-variant pairs in cis window (e.g., $\pm$ 1Mb)

The high-dimensional challenge: $p \gg n$

Predicting expression from genotypes is a high-dimensional problem:

$$Y_i = \sum_{j=1}^{p} X_{ij}\theta_j + \epsilon$$

where θ_j are the **multivariate/joint effects** (cf. β_j for univariate GWAS effects)

- $p \gg n$: need to estimate $\theta_1, \ldots, \theta_p$

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- $p \gg n$: need to estimate $\theta_1, \ldots, \theta_p$
- How many samples? How many unknowns?
- Is it computationally feasible?
- Standard linear regression fails!

Penalized regression: Motivation

- Many predictors (variants), few samples
- Need to select relevant variants
- Avoid overfitting
- Two main approaches:
    - Ridge (L2): shrinkage without variable selection
    - LASSO (L1): shrinkage + sparse selection
    - Elastic Net: combination of both

Application: PrediXcan framework

Goal: Predict gene expression from genotypes to understand GWAS

Two-stage approach:

1. **Training**: Build expression prediction model

$$\hat{E}_g = \sum_j X_j \beta_j^{(g)}$$

   using LASSO/Elastic Net on eQTL reference panel

2. **Prediction**: Test predicted expression with trait

$$Y \sim \hat{E}_g + \text{covariates}$$

Gamazon et al. (2015) Nature Genetics

Lasso: a linear regression with L1 (Tibshirani 1996)

Prior distribution

$$p(\boldsymbol{\theta}) = \mathrm{Laplace}(\boldsymbol{\theta}|\lambda) \propto \exp\left(-\lambda\|\boldsymbol{\theta}\|_1\right)$$

where

$$\|\boldsymbol{\theta}\|_1 = \sum_{j=1}^{p} |\theta_j|, \text{ L1-norm.}$$

Maximize

$$\ln p(\mathbf{y}|X, \boldsymbol{\theta}) + \ln p(\boldsymbol{\theta}|\lambda) = -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \mathbf{x}_i\boldsymbol{\theta})^2 - \lambda\|\boldsymbol{\theta}\|_1$$

Minimize L_1-regularized error

$$\sum_{i=1}^{n}(y_i - \mathbf{x}_i\boldsymbol{\theta})^2 + \lambda\|\boldsymbol{\theta}\|_1$$

`glmnet` (Friedman et al. 2010) solves this opt. problem

Goal (by variable-by-variable updates):

$$\min_{\boldsymbol{\theta}} \quad \overbrace{(\mathbf{y} - X\boldsymbol{\theta})^\top(\mathbf{y} - X\boldsymbol{\theta})}^{\text{RSS}} + \underbrace{\lambda\alpha\|\boldsymbol{\theta}\|_1}_{\text{variable selection}} + \underbrace{\lambda(1-\alpha)\|\boldsymbol{\theta}\|_2}_{\text{shrinkage}}$$

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For each θ_j,

$$\hat{\theta}_j^{\text{glmnet}} \leftarrow \frac{S\left(\sum_{i=1}^n X_{ij}(y_i - \hat{y}_i^{(-j)}), \lambda\alpha\right)}{\sum_{i=1}^n X_{ij}^2 + \lambda(1-\alpha)}$$

Friedman *et al.*, Regularization Paths for Generalized Linear Models via Coordinate Descent (2010)

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For each θ_j,

$$\hat{\theta}_j \leftarrow \frac{\overset{\text{threshold}}{S}\left(\sum_{i=1}^{n} X_{ij} \overbrace{(y_i - y_i^{(-j)})}^{\text{residual w/o the variable } \theta_j}, \lambda\alpha\right)}{\sum_{i=1}^{n} X_{ij}^2 + \underbrace{\lambda(1-\alpha)}_{\text{shrinkage}}}$$

where $S(z, \tau)$ will set it to zero if $|z| < \tau$.

Sum of Single Effect (SuSiE) regression

```r
library(susieR)
Y.train <- .sim$y.q[.train]
susie.out <- susie(X, Y.train)
```

$$\mathbf{y} = \sum_{l=1}^{L} \underbrace{\sum_{j} \mathbf{x}_j \overbrace{\alpha_j^{(l)}}^{\text{probabilistic selection}} \overbrace{\beta_j^{(l)}}^{\text{single variant effect}}}_{\text{layer-by-layer}} + \boldsymbol{\epsilon}$$

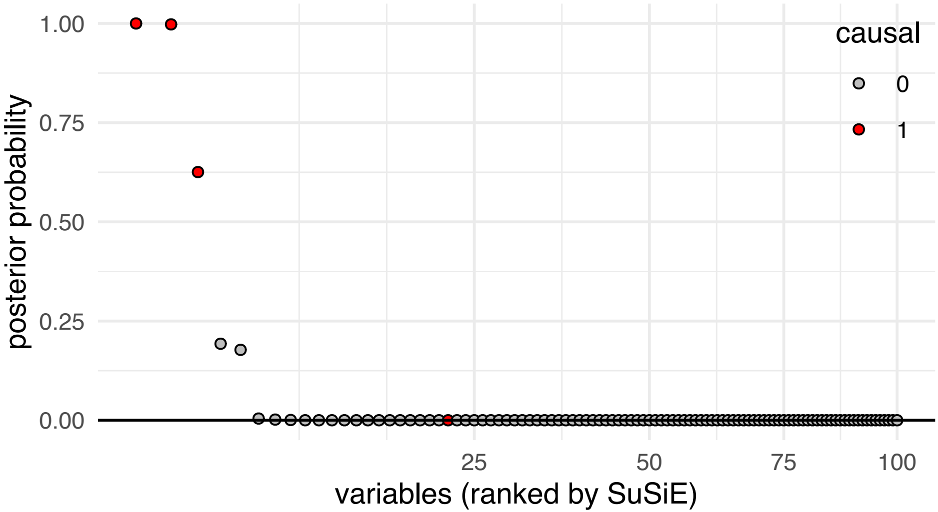
where $\sum_{j=1}^{p} \alpha_j^{(l)} = 1$ for each layer l.

(Wang et al. 2020)

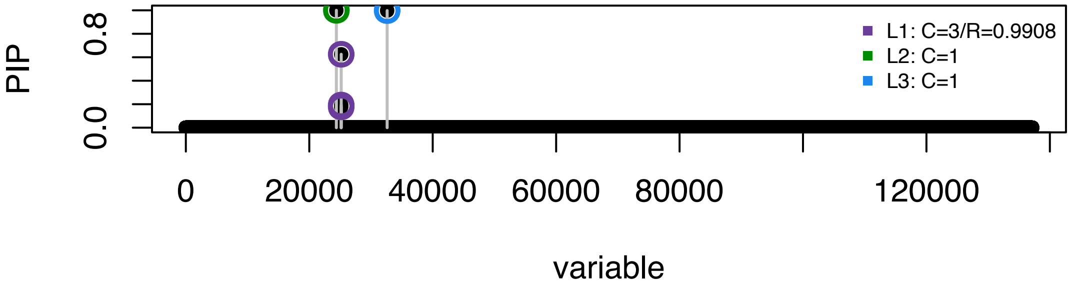
Ideas behind SuSiE

1. Estimating univariate effects is easy (GWAS)
2. Many variants can show similar effects (LD)
3. Let's weight variants probabilistically
4. Regress out the probabilistically reweighted effects
5. Repeat 1-4.

SuSiE identifies top causal variants



SuSiE provides credible sets

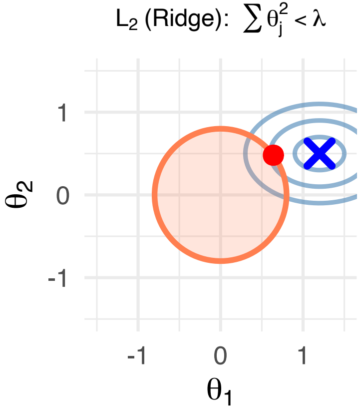


Credible sets: Groups of variants that contain the causal variant with high probability (e.g., 95%)

Each color represents one independent signal (one "layer" in the model)

Geometric intuition: L2, L1, and SuSiE

$$\mathbf{y} = \mathbf{x}_1\theta_1 + \mathbf{x}_2\theta_2 + \epsilon$$



Blue cross ($\times$): OLS solution; Red dot: Constrained solution; Blue contours: RSS levels

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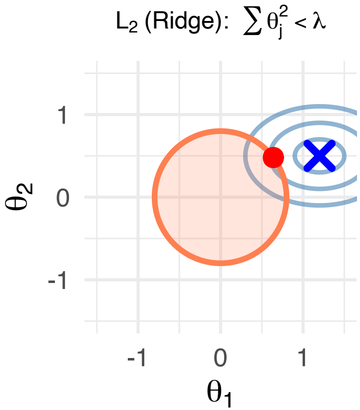

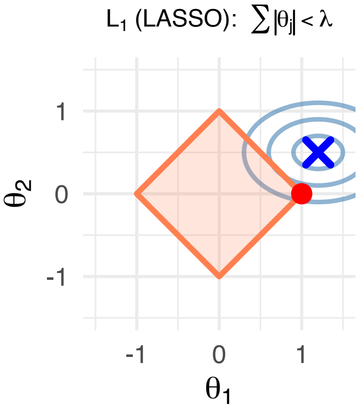

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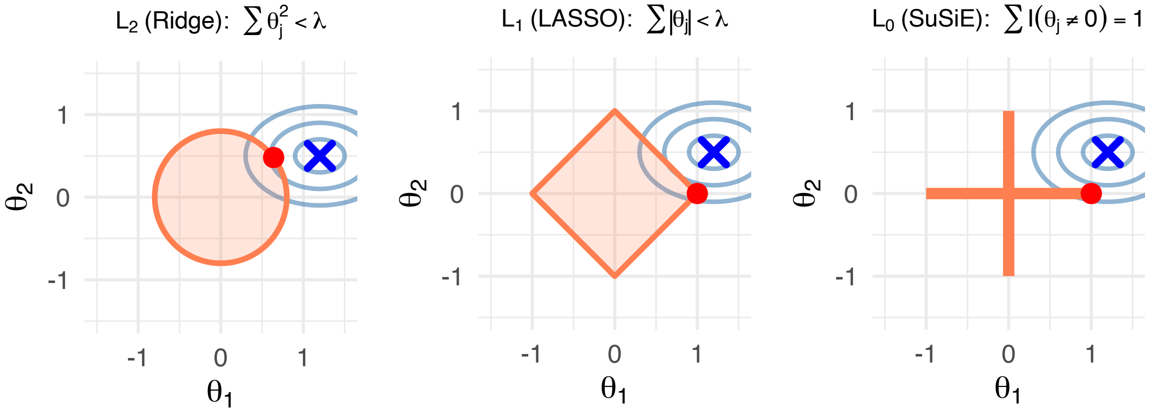

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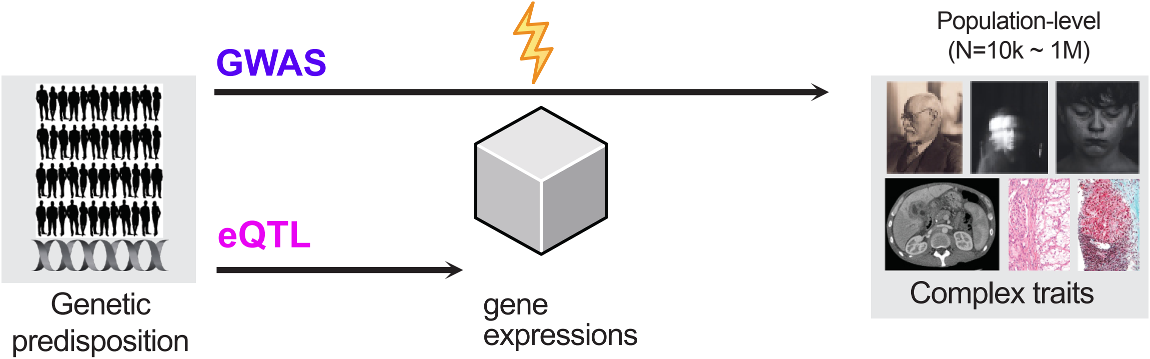

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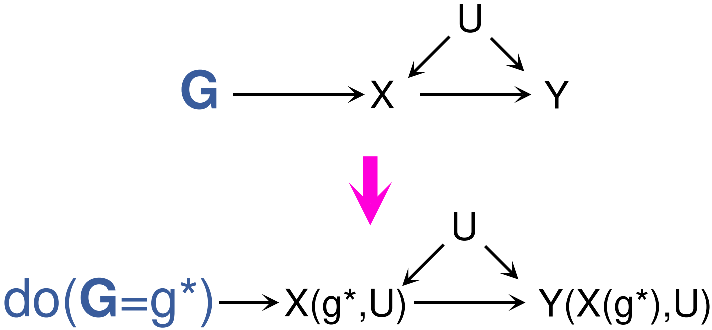
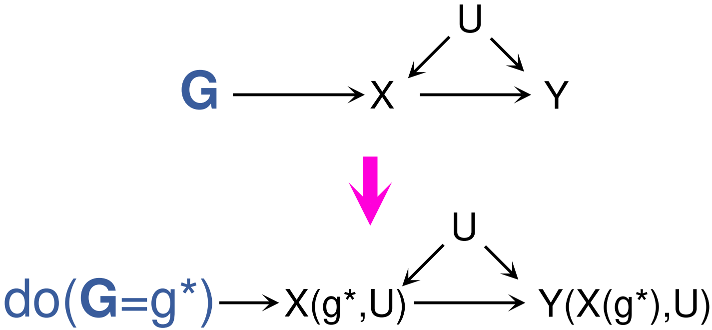
Mendelian Randomization



We will use a genetic variable G to mimic an RCT

Mendelian Randomization



If a genetic variable G is a valid instrumental variable...

MR in modern epidemiology studies

- G: genotype
- X: APOE protein
- Y: cancer

$$G \rightarrow X \rightarrow Y$$

APOLIPOPROTEIN E ISOFORMS, SERUM CHOLESTEROL, AND CANCER

SIR,—It is unclear whether the relation between low serum cholesterol levels and cancer[1] is causal. In many studies occult tumour may have depressed cholesterol levels though in others the relation was found when serum cholesterol had been measured many years before the cancer was diagnosed. The relation is probably not explained by diet, because in the Seven Countries Study cohorts with widely different diets and corresponding differences in mean cholesterol levels experienced similar mean cancer rates.[2,3] On the other hand, within each region cancer incidence was higher in men with a serum cholesterol in the lowest part of the cholesterol distribution for that country.[3] Thus, naturally low cholesterol levels are sometimes associated with increased cancer risk.[1,3]

Differences in the aminoacid sequence of apolipoprotein E (apo

Katan, *Lancet*, (1986)

MR in modern epidemiology studies

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$$G \to X \to Y$$

George
Davey Smith

30TH THOMAS FRANCIS JR MEMORIAL LECTURE

'Mendelian randomization': can genetic epidemiology contribute to understanding environmental determinants of disease?*

George Davey Smith and Shah Ebrahim

APOLIPOPROTEIN E ISOFORMS, SERUM CHOLESTEROL, AND CANCER

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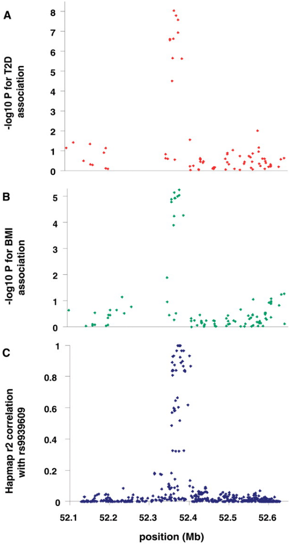
MR in action: *FTO* → fat mass → obesity, diabetes



Using genotype as an instrumental variable, we test causality between other exposure variables and downstream phenotypes.

- Genotype in *FTO* locus → T2D

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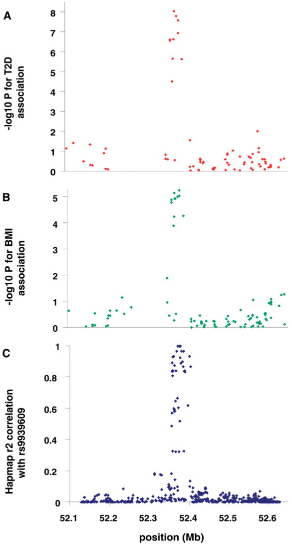

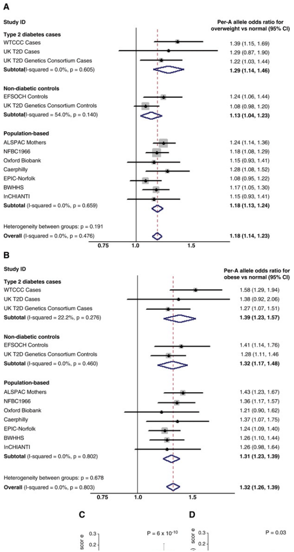

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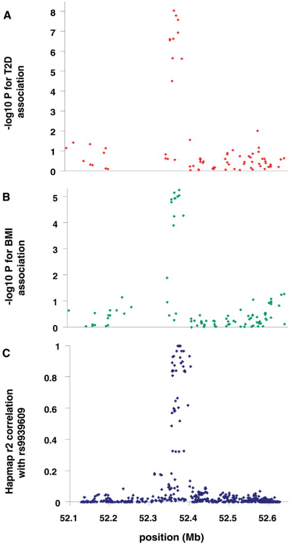
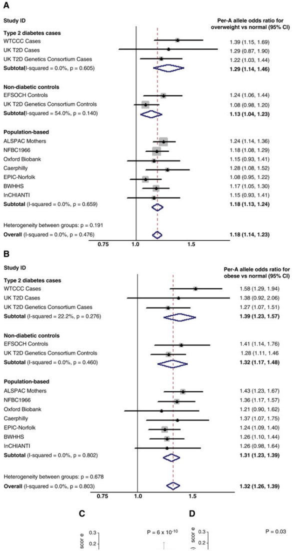
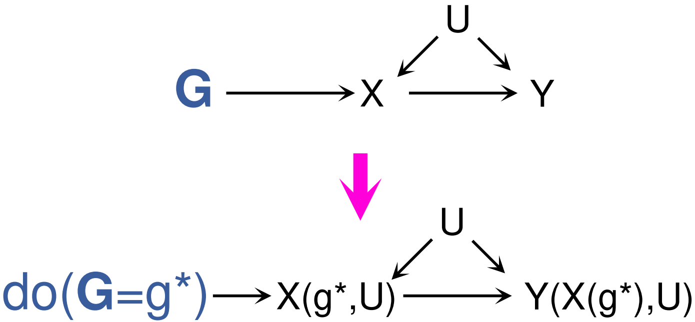




Using genotype as an instrumental variable, we test causality between other exposure variables and downstream phenotypes.

- Genotype in *FTO* locus → T2D
- *FTO* locus → fat mass
- Using *FTO* as "instrumental variable", we can ask other MR questions

Why use Mendelian Randomization?



1. We do not have enough knowledge about U
2. We do not have a way to intervene on X, i.e., $do(X = x)$

"Genotypes are beautifully randomized" (Fisher 1952)

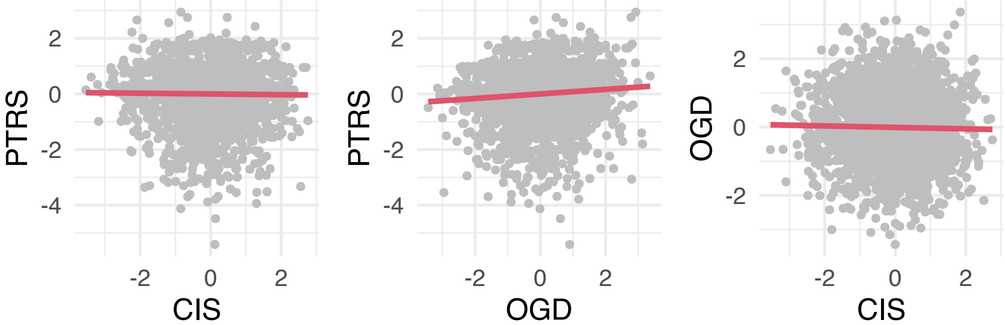
How can we learn causality between genetic disorders?

Three disorders (just for demonstration)

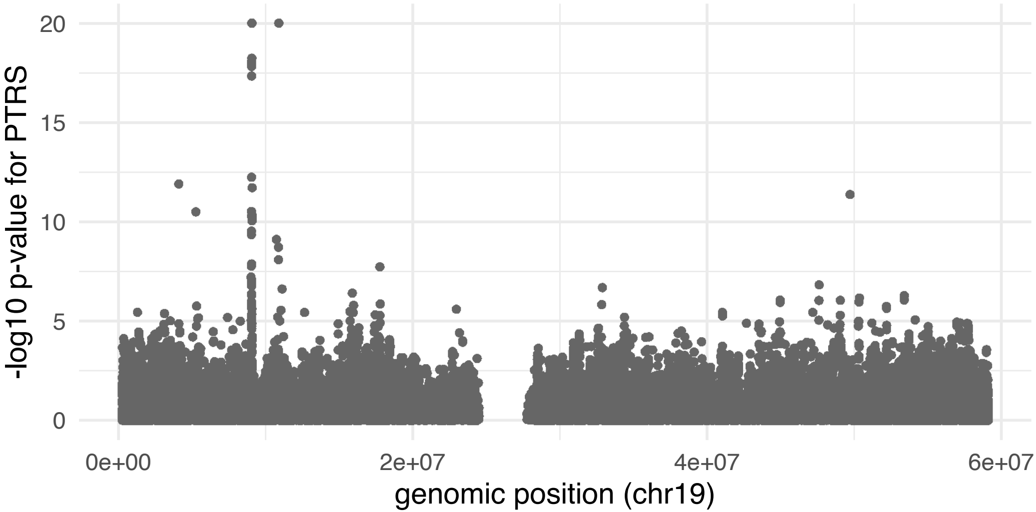
- OGD: Obsessive `ggplot` disorder
- PTRS: Post-traumatic `R` session stress syndrome
- CIS: Chronic indentation syndrome (as a result of `Python` coding)

How can we learn causality between genetic traits?

In previous longitudinal studies, we found the following relationships.



Can we recover such relationships from GWAS data?



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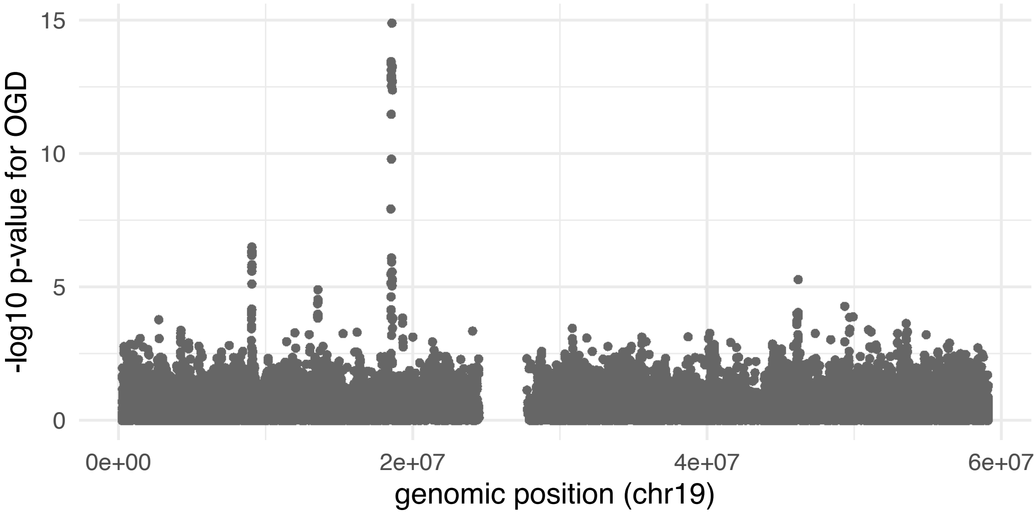

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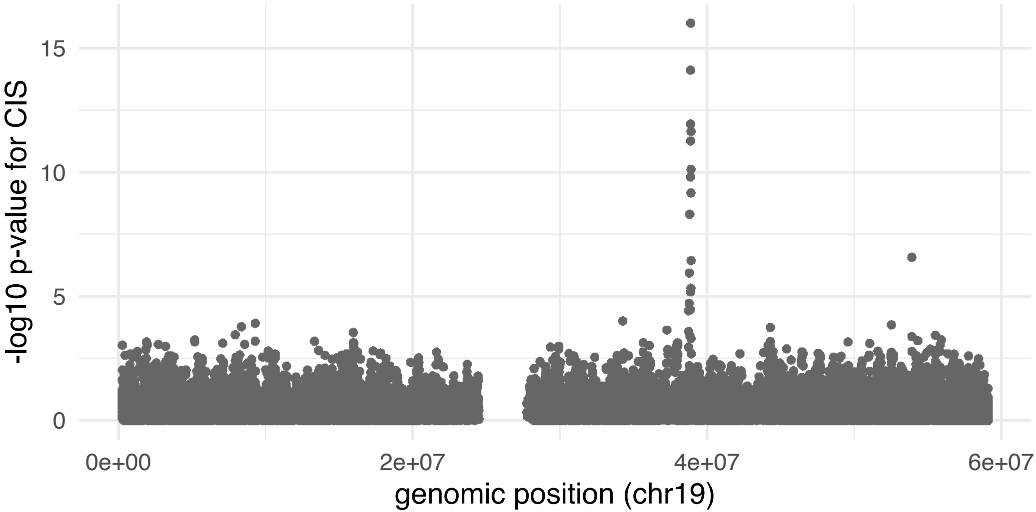

MR measures mediation effects of X to Y

Example:

- X: gene expression
- Y: disease phenotype

Suppose we have estimated

$$\begin{aligned} G &\xrightarrow{\alpha} X \\ X &= G\hat{\boldsymbol{\alpha}} + \epsilon_X \end{aligned}$$

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Goal: What is the causal effect β in $X \xrightarrow{\beta} Y$?

$$Y = X\beta + \epsilon'$$

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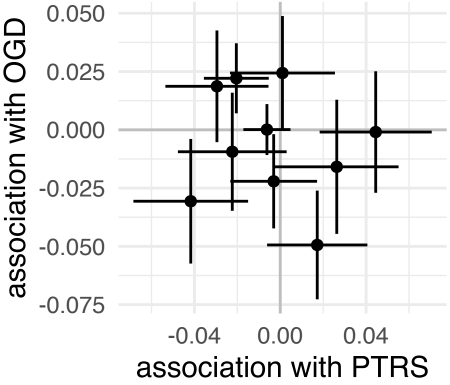
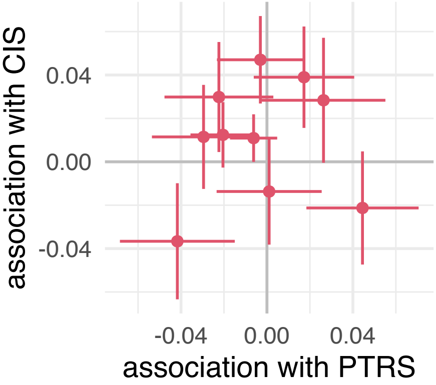
The answer is as simple as

$$\mathbb{E}[\beta] = \frac{\boldsymbol{\gamma}}{\boldsymbol{\alpha}}$$

Will any genetic variable work for MR analysis?

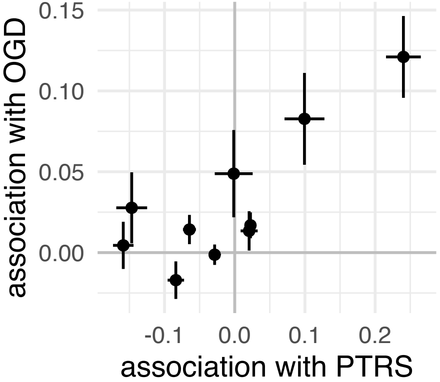
Goal: Is PTRS $\rightarrow$ OGD or PTRS $\rightarrow$ CIS?

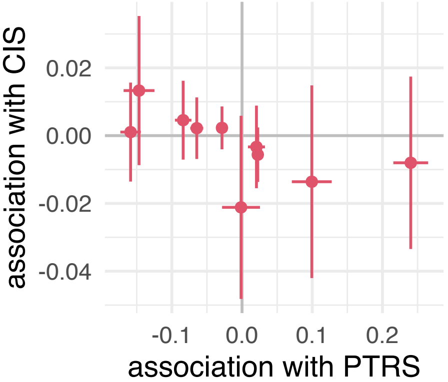




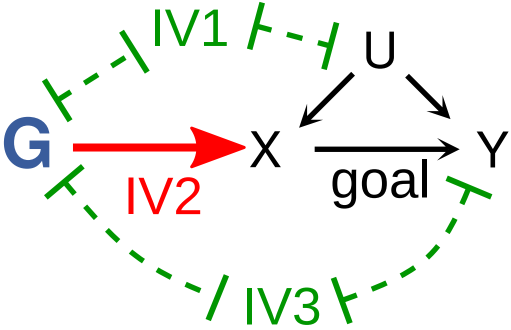
Select causal variants of exposure (PTRS)?

Goal: Is PTRS $\rightarrow$ OGD or PTRS $\rightarrow$ CIS?





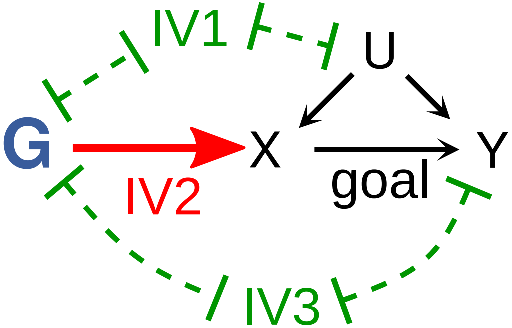
Definition of instrumental variable in MR study



(Bowden et al. 2015)

- IV1: The genetic variant G is independent of the potential confounder variable U

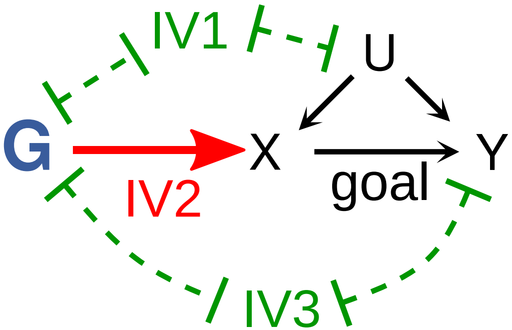
Definition of instrumental variable in MR study



- IV1: The genetic variant G is independent of the potential confounder variable U
- IV2: The genetic variant is associated with the exposure X

(Bowden et al. 2015)

Definition of instrumental variable in MR study



- IV1: The genetic variant G is independent of the potential confounder variable U
- IV2: The genetic variant is associated with the exposure X
- IV3: The genetic variant is independent of the outcome Y conditioning on X

(Bowden et al. 2015)

Today's lecture

1. Polygenic risk prediction (G $\rightarrow$ phenotype)
2. Fundamental challenges in PGS (regression)
3. Expression Quantitative Trait Loci (G $\rightarrow$ Expr.)
4. Co-localization in genomics

What is co-localization?

Co-localization: Two traits share the same causal variant(s) in a genomic region

- If two GWAS signals co-localize, they likely share a causal variant
- Important for: MR validation, functional follow-up, understanding pleiotropy

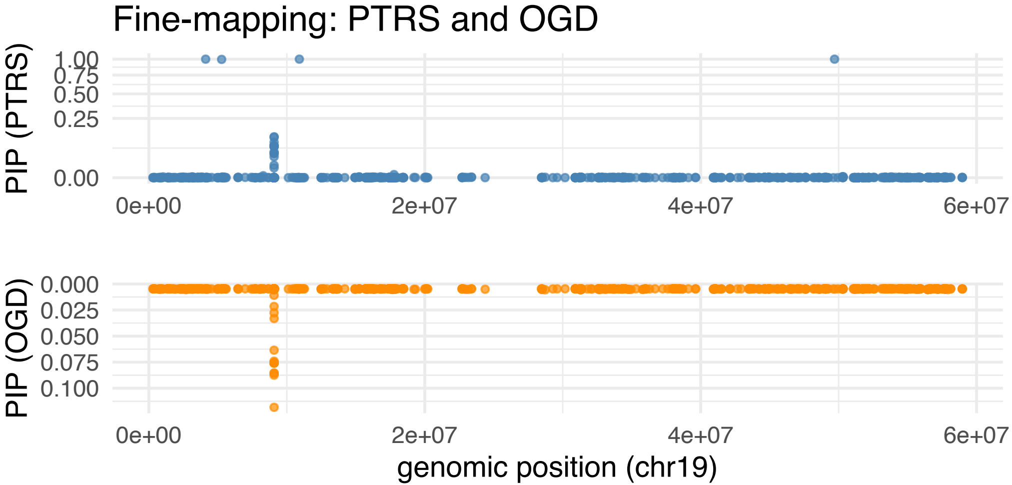
Our simulation:

- PTRS and OGD share 2 causal variants (co-localized)

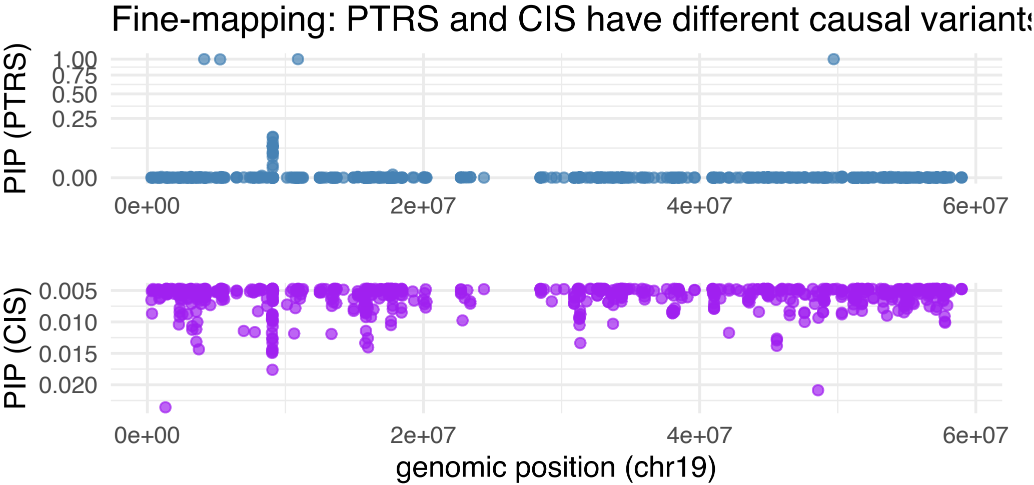
Fine-mapping with SuSiE

- SuSiE (Sum of Single Effects) identifies causal variants
- Outputs PIP (Posterior Inclusion Probability) for each variant
- High PIP $=$ likely to be causal

Co-localization: PTRS and OGD (mediated)



No co-localization: PTRS and CIS (independent)



`coloc.susie` compare two SuSiE results

Posterior probabilities (PP) (Wallace 2021):

- **PP.H0**: No association in either trait
- **PP.H1**: Association with trait 1 only
- **PP.H2**: Association with trait 2 only
- **PP.H3**: Both associated, different causal variants
- **PP.H4**: Both associated, **shared causal variant** (co-localization)

Rule of thumb: PP.H4 > 0.8 suggests strong evidence for co-localization

coloc.susie allows for multiple causal variants per trait

Statistical test: coloc with SuSiE

Using coloc.susie (Wallace 2021) to test for co-localization:

```
## PTRS vs OGD (should co-localize)
coloc.ptrs.ogd$summary
```

```
##    nsnps     hit1    hit2     PP.H0.abf    PP.H1.abf     PP.H2.abf PP.H3.abf
##    <int>   <char>  <char>         <num>        <num>         <num>     <num>
## 1:   841 10907286 9078190 1.688239e-21 0.0211344301 7.817507e-20 0.9786441
## 2:   841 49705291 9078190 2.045804e-06 0.0211225288 9.473239e-05 0.9780921
## 3:   841  5273571 9078190 1.436645e-04 0.0209848186 6.652486e-03 0.9717157
## 4:   841  4115684 9078190 7.781789e-08 0.0211264781 3.603412e-06 0.9782751
## 5:   841  9079464 9078190 9.825459e-22 0.0006701868 4.549747e-20 0.0290930
##       PP.H4.abf  idx1  idx2
##           <num> <int> <int>
## 1: 0.0002214617     1     1
## 2: 0.0006886163     3     1
## 3: 0.0005033485     4     1
## 4: 0.0005947005     5     1
## 5: 0.9702368166     2     1
```

Statistical test: coloc with SuSiE

Using coloc.susie (Wallace 2021) to test for co-localization:

```
## PTRS vs CIS (should NOT co-localize)
coloc.ptrs.cis$summary
```

```
## NULL
```

Summary: Visual + Statistical evidence

Co-localized (PTRS-OGD):

- High PIPs at same positions (mirror image)
- High PP.H4 from coloc.susie
- Suggests shared causal variant

Not co-localized (PTRS-CIS):

- High PIPs at different positions
- Low PP.H4, higher PP.H3
- Independent causal variants

Working with GWAS summary statistics

Often we only have access to GWAS summary statistics, not individual-level data:

- $\hat{\beta}_j$: effect size for variant j
- $\mathrm{SE}(\hat{\beta}_j)$: standard error
- $Z_j = \hat{\beta}_j / \mathrm{SE}(\hat{\beta}_j)$: z-score
- $\chi_j^2 = Z_j^2$: chi-square statistic

Question: Can we infer biological mechanisms from summary statistics alone?

From univariate to multivariate effects

Underlying assumption: traits are polygenic

$$Y_i = \sum_{j=1}^{p} X_{ij}\theta_j + \epsilon$$

Problem: Univariate GWAS gives us $\hat{\beta}_j$ for each variant j separately

Goal: Understand the true multivariate effects θ_j from summary statistics

Challenge: LD structure confounds the relationship between $\hat{\beta}_j$ and θ_j

LD-score: Relating χ^2 statistics to heritability

What is a generative model for χ_j^2 statistics?

We can show that the univariate z-score relates to multivariate effects:

$$Z_j = \frac{\sqrt{n}}{\sigma} \sum_k R_{jk} \theta_k + \epsilon_j$$

where R is the LD matrix and $\epsilon \sim \mathcal{N}(0, 1)$.

Bulik-Sullivan *et al.*, *Nature Genetics* (2014)

LD-score: Relating χ^2 statistics to heritability

What is a generative model for χ_j^2 statistics?

$$\mathbb{E}\left[\chi_j^2\right] = \mathbb{E}\left[Z_j^2\right] = \mathbb{E}\left(\sqrt{n}\sum_k R_{jk}\theta_k + \epsilon_j\right)^2$$

Bulik-Sullivan *et al.*, *Nature Genetics* (2014)

LD-score: Relating χ^2 statistics to heritability

What is a generative model for χ^2_j statistics?

$$\mathbb{E}\left[\chi^2_j\right] = \mathbb{E}\left[Z^2_j\right] = \mathbb{E}\left(\sqrt{n}\sum_k R_{jk}\theta_k + \epsilon_j\right)^2$$

If the effects are independent, i.e., $\mathbb{E}[\theta_k\theta_j] = 0$ for all $k \neq j$,

$$\mathbb{E}\left[\chi^2_j\right] = n \underbrace{\sum_k R^2_{jk}}_{\text{LD-score}} \mathbb{E}\left[\theta^2_k\right] + 1$$

Bulik-Sullivan *et al.*, *Nature Genetics* (2014)

Baseline LD-score to measure polygenic heritability

Assuming that all the variants equally contribute,

$$\mathbb{E}\left[\theta_k^2\right] = \tau / p,$$

where p is the total number of SNPs.

Defining an LD score for a variant/SNP j as

$$l_j \overset{\text{def}}{=} \sum_k R_{jk}^2,$$

We get

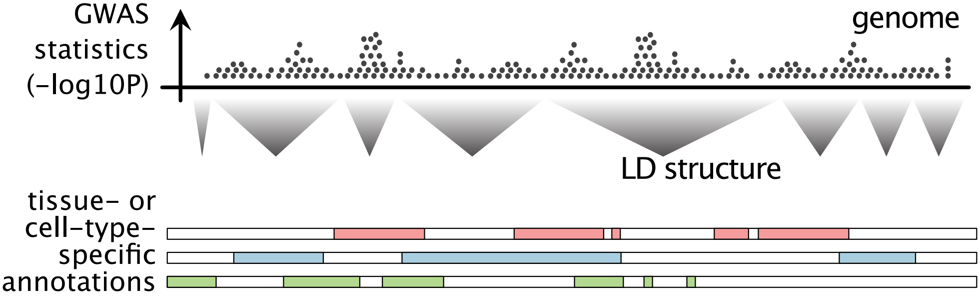
$$\mathbb{E}\left[\chi_j^2\right] = \underbrace{n}_{\text{sample size}} \underbrace{l_j}_{\text{LD score}} \underbrace{\frac{\tau}{p}}_{\text{per SNP heritability}} + 1$$

Regression model: Regress observed χ_j^2 on LD scores l_j to estimate τ (heritability)

Bulik-Sullivan *et al.*, *Nature Genetics* (2014)

Stratified LD-score regression (S-LDSC)

Question: Which genomic annotations explain GWAS heritability?



(Finucane, Bulik-Sullivan, Gusev, Trynka, Reshef, Loh, Anttila, Xu, Zang, Farh, and others 2015)

Stratified LD-score regression (S-LDSC)

Question: Which genomic annotations explain GWAS heritability?

Partition heritability across C genomic annotations:

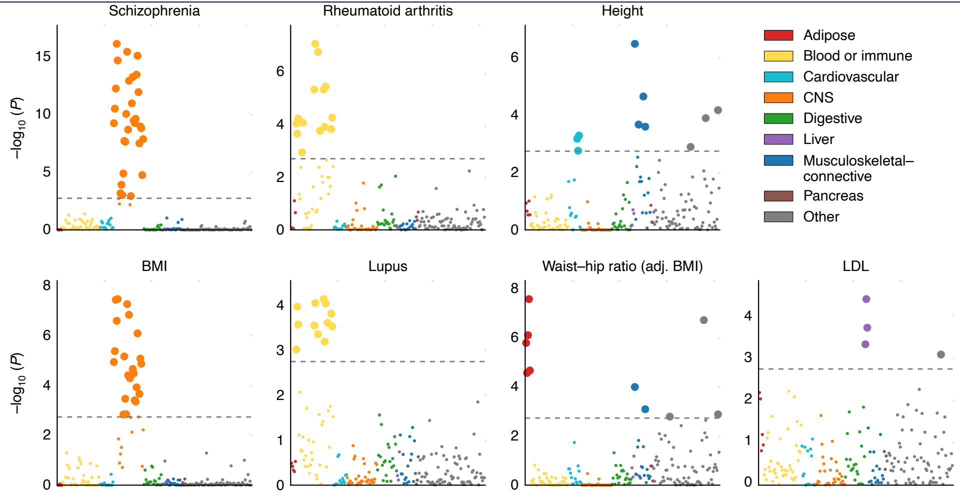
$$\mathbb{E}\left[\chi_j^2\right] = n \sum_{c=1}^{C} \ell_{j,c} \tau_c + 1$$

where:

- $\ell_{j,c}$: LD score for variant j with respect to annotation c - τ_c: per-SNP heritability in annotation c

(Finucane, Bulik-Sullivan, Gusev, Trynka, Reshef, Loh, Anttila, Xu, Zang, Farh, and others 2015)

S-LDSC → tissue/cell type contexts of GWAS



(Finucane, Bulik-Sullivan, Gusev, Trynka, Reshef, Loh, Anttila, Xu, Zang, Farh, Ripke, et al. 2015)

Take-home messages

1. Methods like SuSiE, coloc, and TWAS integrate multiple data types
2. LD score regression connects GWAS signals to heritability
3. Regression is fundamental to dissecting regulatory genomic mechanisms

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